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Acoustic-Based Detection, Classification, and Localization of Impulsive Sounds

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Motivation and research questions

- Acoustic detection of gunfire complements camera-based surveillance: the acoustic channel is omnidirectional and does not require line of sight.
- Three required functions, in outdoor environments that are noisy and reverberant:



detection

of the impulsive event



classification

ideally down to calibre



localization

of the source, in 2D

This study

1. Which cepstral representation separates gunshots from acoustically similar false alarms most reliably?
2. Neural network or support vector machine on those features?
3. How does the analysis frame length interact with environmental reflections?

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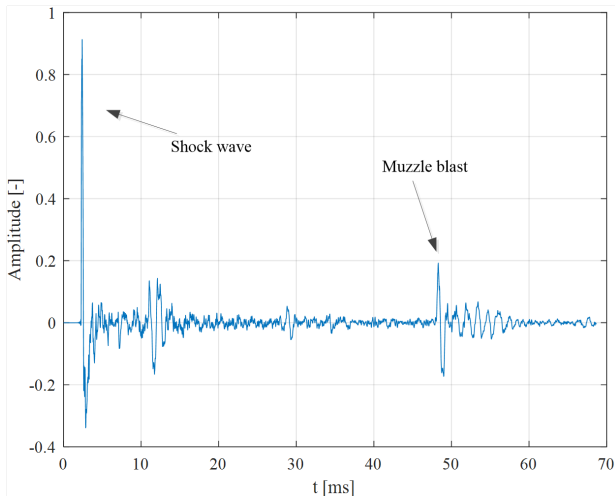
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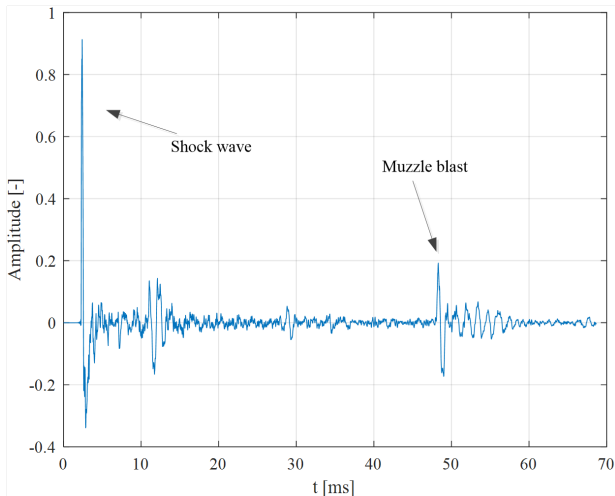
The acoustic signature of a gunshot



One or two components, depending on projectile speed:

- **Shock wave** (“N-wave”) – *supersonic only* ($v > c$); generated along the trajectory, so it arrives first.
- **Muzzle blast** – *always present*; propellant gases at the barrel, 1–3 ms; here ≈ 45 ms behind.
- Between and after: **reflections**.

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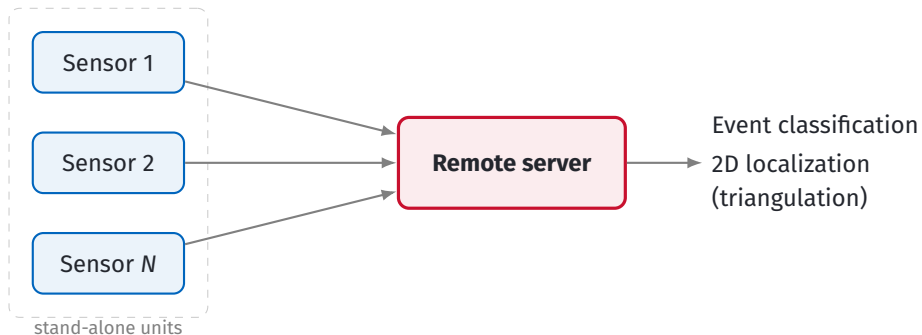
Single-domain analysis is insufficient

- **Spectral analysis alone** often captures environmental noise rather than the gunshot itself.
- **Time-domain patterns** get distorted by nonlinear dispersion and obstacles.

Consequence

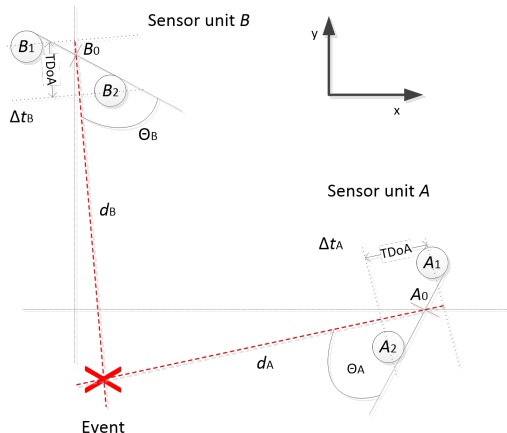
Effective feature extraction must integrate *both* time and frequency domain processing.

Proposed system architecture



- Each sensor continuously monitors its surroundings and transmits detected signals to the server for advanced processing.
- Multiple units give broader coverage and adaptability to environments.
- Tested with various firearm calibers and ammunition.

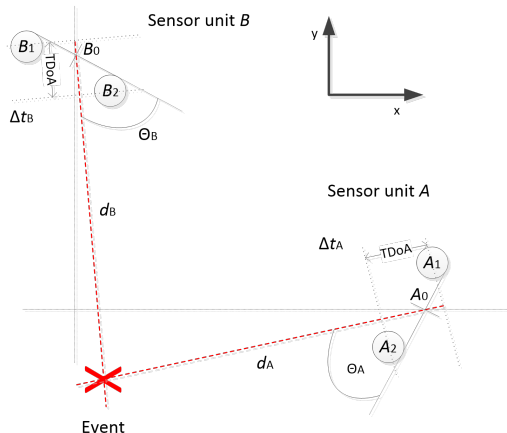
Localization – bearing from TDoA



$$\Theta = \arcsin\left(\frac{c \Delta n}{d f_s}\right)$$

- Two MEMS microphones per unit ($d = 0.186$ m, $f_s = 44.1$ kHz) give one **bearing** Θ .
- Two units, two bearings – they **intersect** at the event.
- Measured: cross-correlation with parabolic interpolation, **MAE 0.77°** over 70 events (-45° to $+60^\circ$).

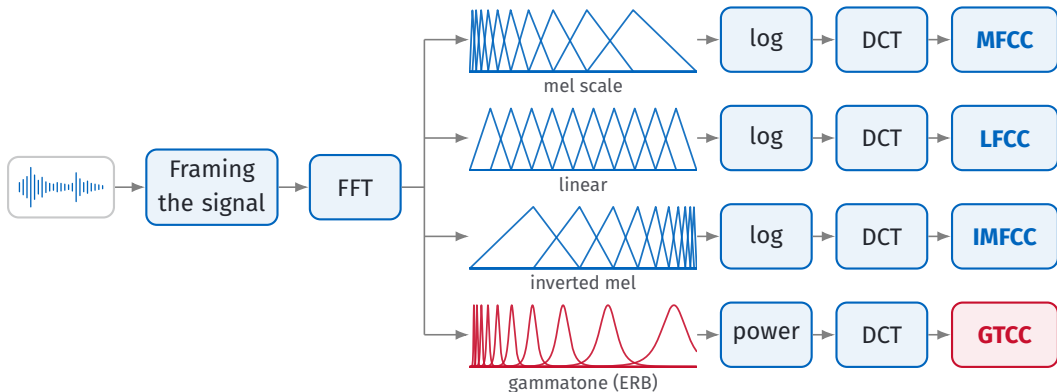
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Feature extraction – cepstral coefficients



Feature extraction – the four filter banks



MFCC – mel scale

dense at low frequencies;
speech-born,
validated on
gunshots at
military ranges



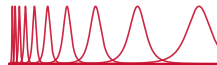
LFCC – linear

equally spaced
filters, uniform
resolution
across the band



IMFCC –

inverted mel
mirror of the mel
scale: dense at
high frequencies



GTCC – gammatone

auditory model, ERB
spacing; **reported**
more noise-robust
than MFCC

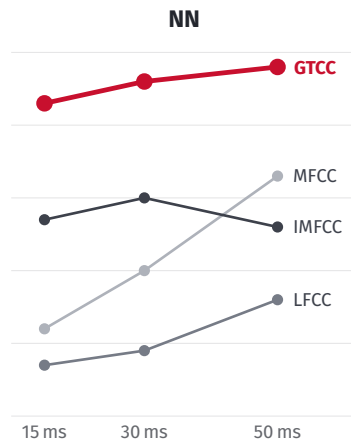
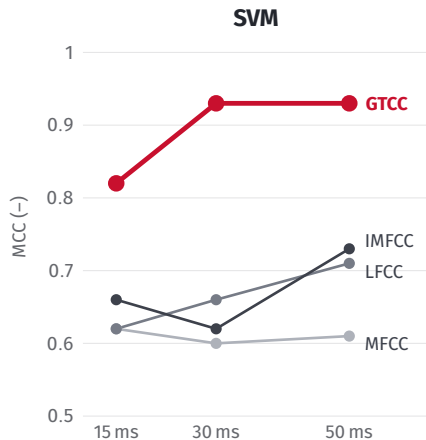
Experimental grid: 4 extractors $\times$ 3 frame lengths (15 / 30 / 50 ms) $\times$ 2 classifiers (NN, SVM).

Classification results – 9 mm class

Frame	Extractor	ACC (%)		RLC (%)		MCC (–)	
		SVM	NN	SVM	NN	SVM	NN
15 ms	MFCC	83.90	85.59	80.65	67.74	0.62	0.62
	LFCC	83.90	83.90	80.65	64.52	0.62	0.57
	IMFCC	85.59	90.68	83.87	87.10	0.66	0.77
	GTCC	93.22	97.46	87.10	93.55	0.82	0.93
30 ms	MFCC	84.75	87.29	67.74	87.10	0.60	0.70
	LFCC	87.29	83.90	70.97	70.97	0.66	0.59
	IMFCC	85.59	92.37	67.74	80.65	0.62	0.80
	GTCC	97.46	98.31	90.32	93.55	0.93	0.96
50 ms	MFCC	84.75	93.22	74.19	90.32	0.61	0.83
	LFCC	88.98	86.44	77.42	77.42	0.71	0.66
	IMFCC	89.83	90.68	74.19	80.65	0.73	0.76
	GTCC	97.46	99.15	90.32	96.77	0.93	0.98

Classification results for the 9 mm class. Test partition: 118 events, 31 of them 9 mm.

Classification results – MCC at a glance



MCC for the 9 mm class by frame length – the same data as the table (118 test events, 31 of them 9 mm).

- **MFCC, LFCC and IMFCC yield similar results.**
- **GTCC consistently performs better** across every frame length and both classifiers.
- **NN outperforms SVM** – better at modelling complex patterns, especially with longer frames.
- **Longer frames improve accuracy**, but start to capture environmental reflections, which can pull focus away from the gunshot itself.

Best configuration in the table

GTCC + NN at 50 ms – 99.15 % ACC, 96.77 % RLC, MCC 0.98.

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- A distributed network of stand-alone acoustic sensors with a remote server detects, classifies and localizes impulsive sounds in 2D.
- Cepstral coefficients are an effective feature extractor for gunshot classification with both NN and SVM.
- GTCC beats Mel-frequency based coefficients; NN achieves slightly better classification success, especially with longer frames.
- Validated with various firearm calibers and ammunition.

Thank you – questions?

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Appendix – experimental setup and dataset

- Military shooting range, fixed sensor position, clear line of sight to every firing position.
- Four weapons: 9 mm, 5.56 NATO SD, 7.62 mm Tokarev, .22 – each with a single ammunition type.
- Plus a false-alarm class with gunshot-like impulsive patterns.
- 26 features per extractor; $\approx 75 / 25$ train–test split.
- NN: MATLAB Neural Fitting, Levenberg–Marquardt, 370 iterations.

Class		Train	Test
0	False alarms	86	21
1	9 mm	70	31
2	5.56 NATO SD	100	31
3	7.62 mm Tokarev	55	17
4	.22	60	18
Total		371	118

Svatoš & Holub, *Meas. Sci. Technol.* **35** (2024) 076122, Table 2.

Appendix – bearing estimation methods

Method	MAE	Median	Max
Cross-correlation	0.84°	0.41°	7.61°
Cross-correlation + parabolic interp.	0.77°	0.56°	6.67°

70 events, 9 reference angles from -45° to $+60^\circ$, $d = 0.186$ m, $f_s = 44.1$ kHz. A phase-based delay estimator was also evaluated and performed worse (MAE 3.24°); cross-correlation was adopted.

Sub-sample interpolation matters

- max. delay ≈ 24 samples for this geometry
- parabolic fit on the correlation peak
- $0.84^\circ \rightarrow 0.77^\circ$ MAE

How many units for 2D?

- ToA, 1 mic per unit – 3 units
- TDoA, 1 mic per unit – 4 units
- **TDoA, 2 mics per unit – 2 units**

Appendix – other cepstral coefficients

- **LPCC** (Linear Prediction Cepstral Coefficients) and **PLPC** (Perceptual Linear Prediction Coefficients) – noise elimination, speech recognition.
- **GFCC** (Greenwood Function Cepstral Coefficients) – derived from human auditory response models, used for classifying animal sounds.
- **GTCC** (Gammatone-Frequency Cepstral Coefficients) – same family, greater resistance to noise than MFCC.

ACC Accuracy – share of correctly classified events.

RLC Recall, also known as *sensitivity* – share of actual gunshots of the class that were detected.

MCC Matthews Correlation Coefficient – balanced measure robust to class imbalance; +1 perfect, 0 random.

Definitions as used in the source study: Svatoš & Holub, *Meas. Sci. Technol.* **35** (2024) 076122, eqs. (1)–(3).